



SIH 26041 • GOVERNMENT OF JHARKHAND • FINAL BUILD PROPOSAL

SurakshaAR

AR-Based Vocational Safety Training Platform

“Real Hazards. Real Decisions. Safer Tomorrow.” A complete proposal for the final version of the augmented-reality safety trainer for Jharkhand’s mining & manufacturing workforce: vision, technology stack, internal engineering architecture, end-to-end workflows, and delivery roadmap.

Android • Kotlin • ARCore

ASP.NET Core 8 • MongoDB

Unity 6 • AR Foundation

Offline-First • 7 Languages

QR Certificates • Hash-Chain

5

SAFETY DOMAINS

3

USER ROLES

7

LANGUAGES

100%

OFFLINE CAPABLE

PROPOSAL V1.0 • FINAL VERSION

SEPTEMBER 2026

WORKSPACE: E:\SAFE

The Final App – What We Ship

THE PROBLEM

Jharkhand recorded **48 fatal mine accidents** in 2022–23, many involving workers with under 30 days of orientation. Classroom retention drops below 20% in one week. Live drills disrupt production. VR headsets don't reach small mines. No regional-language digital safety platform exists.

- Young tribal recruits, limited digital literacy
- Mid-range Android phones (10+), poor connectivity
- Hindi & Santali mandatory; 7 languages total

THE FINAL SOLUTION

A phone-first **AR safety simulator**: the worker scans a QR checkpoint, a 3D hazard (fire, gas leak) appears in the real workspace, a live HUD demands action, inaction escalates danger, and a scored assessment issues a verifiable certificate – fully usable offline.

- 3-role ecosystem: Worker • Manager • Admin
- One backend, one mobile client, one web dashboard
- Trust chain: signed QR certificates, public verify

FINAL PRODUCT PILLARS

01 • Real AR Training

Plane detection, anchored 3D hazards, HUD telemetry, evacuation arrows – real camera, real risk response.

02 • Escalation Engine

Inaction worsens the scene: Normal • Minor • Moderate • Severe • Critical with score penalties & auto-fail.

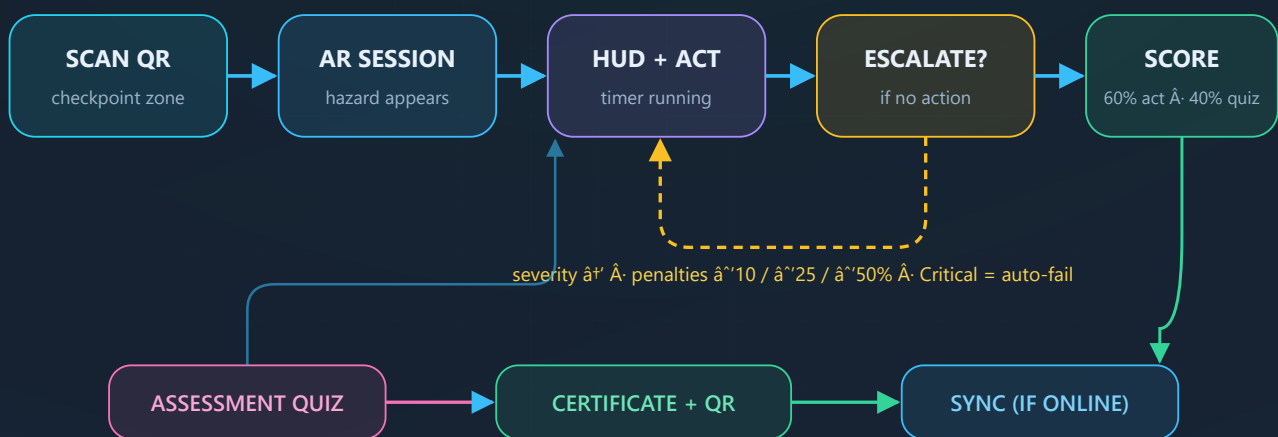
03 • Verifiable Proof

60/40 action+quiz scoring, signed certificates, QR payloads, hash-chain ledger, offline verification.

04 • Offline-First

Full training without internet; queued sync of results & escalations when the network returns.

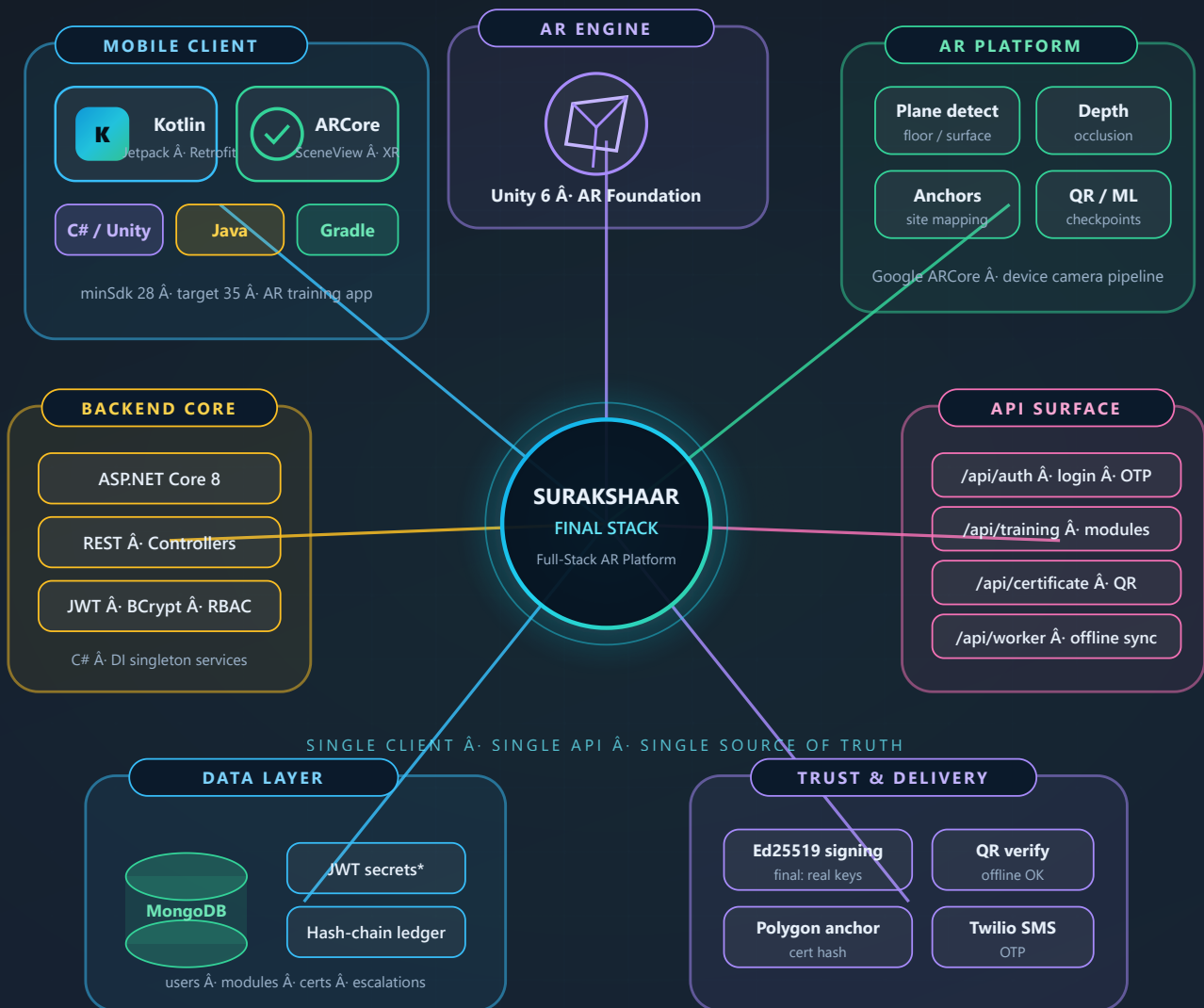
TARGET EXPERIENCE – FINAL APP LOOP



Full loop works with zero connectivity after first cache • results flush via outbox on reconnect

Internal Coding & Technology Stack

FULL-STACK AR PRODUCT



* secrets move to environment / key-vault in final build

End-to-End Platform Architecture

FIELD · DEVICE · API · DATA

1 · FIELD ENVIRONMENT

Coal mine / steel plant
training floor

QR checkpoint
physical zone marker

Safety officer
supervises session

Worker + phone
Android 10+, ARM64

2 · WORKER DEVICE · SurakshaAR APP

PRESENTATION

10 Activities (Kotlin)
Role-based home
HUD · Assessment UI
7-language TTS
TalkBack accessible

DOMAIN / TRAINING

Scenario player
Escalation engine
Scoring 60 / 40
Certificate builder
port of Unity logic

AR RUNTIME

ARCore session
Plane + anchor map
3D hazard prefabs
Evacuation arrows
SceneView / Filament

DATA (LOCAL)

Room DB (final)
Sync outbox queue
Module cache
Cert wallet (JSON)
encrypted prefs

HTTPS · REST/JSON · JWT bearer (when online)

3 · BACKEND · ASP.NET CORE 8 (:5000)

Gateway

CORS · Swagger(dev)
JWT middleware
Static admin SPA
/health

Controllers (7)

Auth · Admin · Manager
Worker · Training
Assessment · Certificate
policy-guarded routes

Services (7)

Auth · Mongo · OTP
Scenario · Escalation
Blockchain · SMS
singleton DI

Shared models

SurakshaAR.Shared
20 entities / DTOs
single contract
C# source of truth

4 · PRIMARY DATA

MongoDB

10 collections · indexed

blockchain coll.

SHA-256 hash chain

5 · EXTERNAL / SIDE CHANNELS

Twilio

OTP SMS

Polygon

cert hash anchor

Admin SPA

:5000/ root

ONLINE PATH

App · REST with JWT; every write lands in Mongo; certificates & escalations also appended to the hash chain.

OFFLINE PATH

Training runs from cached modules; results & escalations queue in an outbox and flush via **/worker/online/sync** on reconnect.

VERIFY PATH

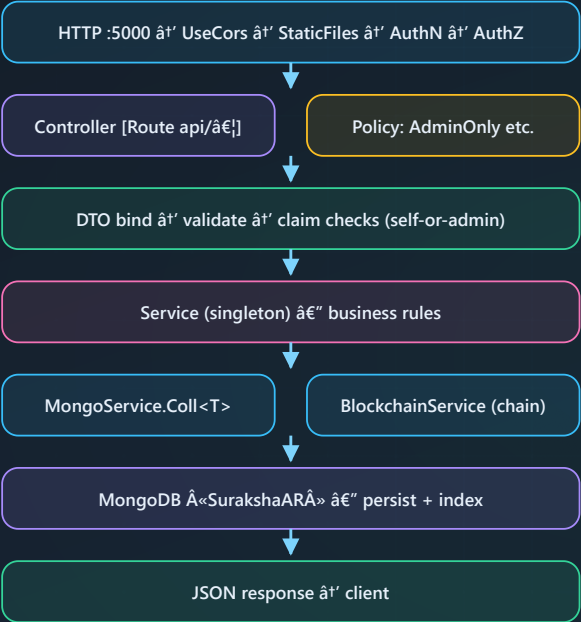
Anyone scans a certificate QR · **/certificate/verify** (anonymous) · validity, score, module · no login required.

Inside the Code “ Backend & App

LAYERED · FEATURE-ALIGNED

BACKEND INTERNAL LAYERING (C# / .NET 8)

REQUEST PIPELINE

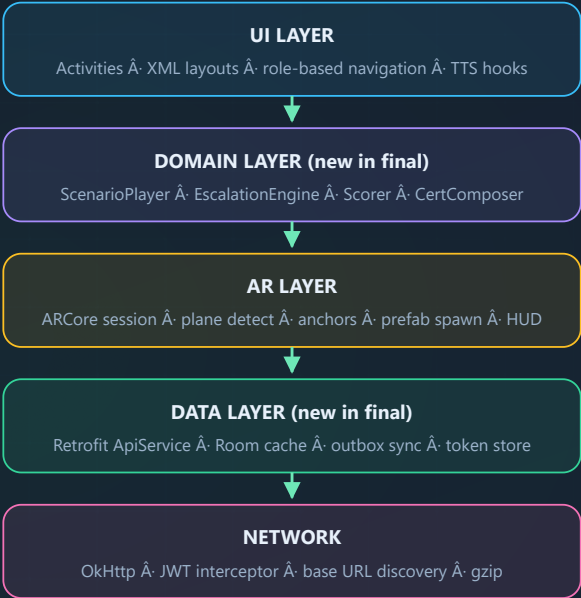


PROJECT MAP

Backend/
Program.cs · pipeline, DI, JWT, policies
Controllers/ · 7 · HTTP surface
Services/ · 7 · Auth Mongo OTP Scenario
Escalation Blockchain Sms
wwwroot/admin/ · vanilla-JS dashboard
Config/MongoConfig · appsettings.json
Shared/Models.cs · 20 entities (single contract)

MOBILE APP INTERNAL LAYERING (KOTLIN “ FINAL)

FINAL RUNTIME LAYERS



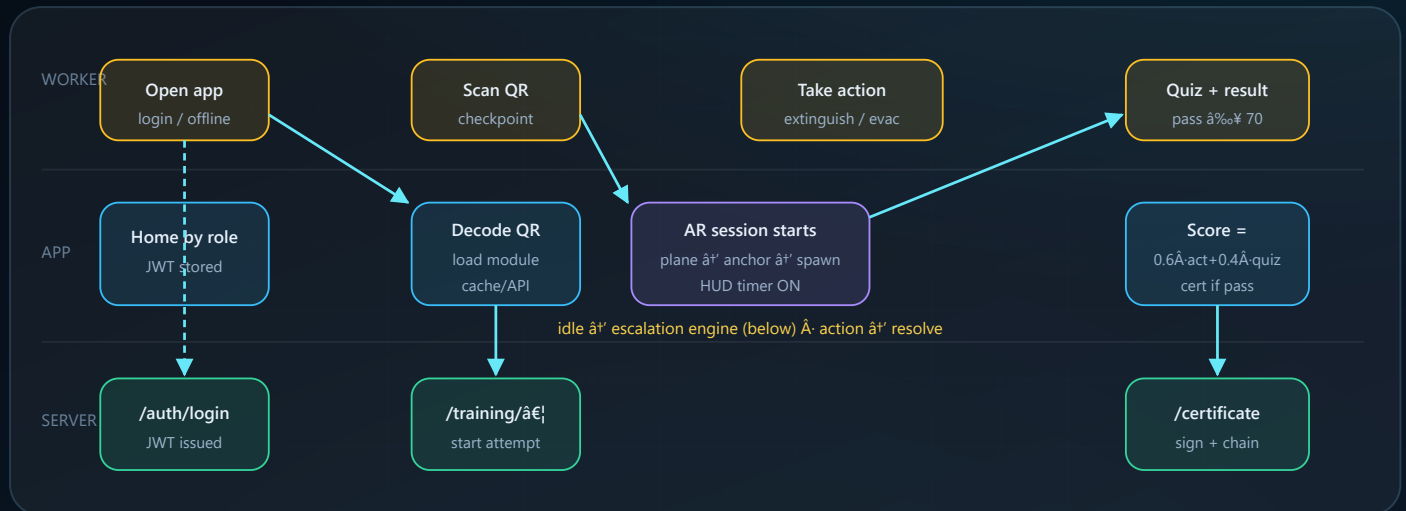
FINAL BUILD DECISIONS

- One client “ kill dual-frontend drift (native wins)
- Port Unity training logic · Kotlin domain layer
- Room outbox for true offline sync
- Shared models mirrored via codegen checklist

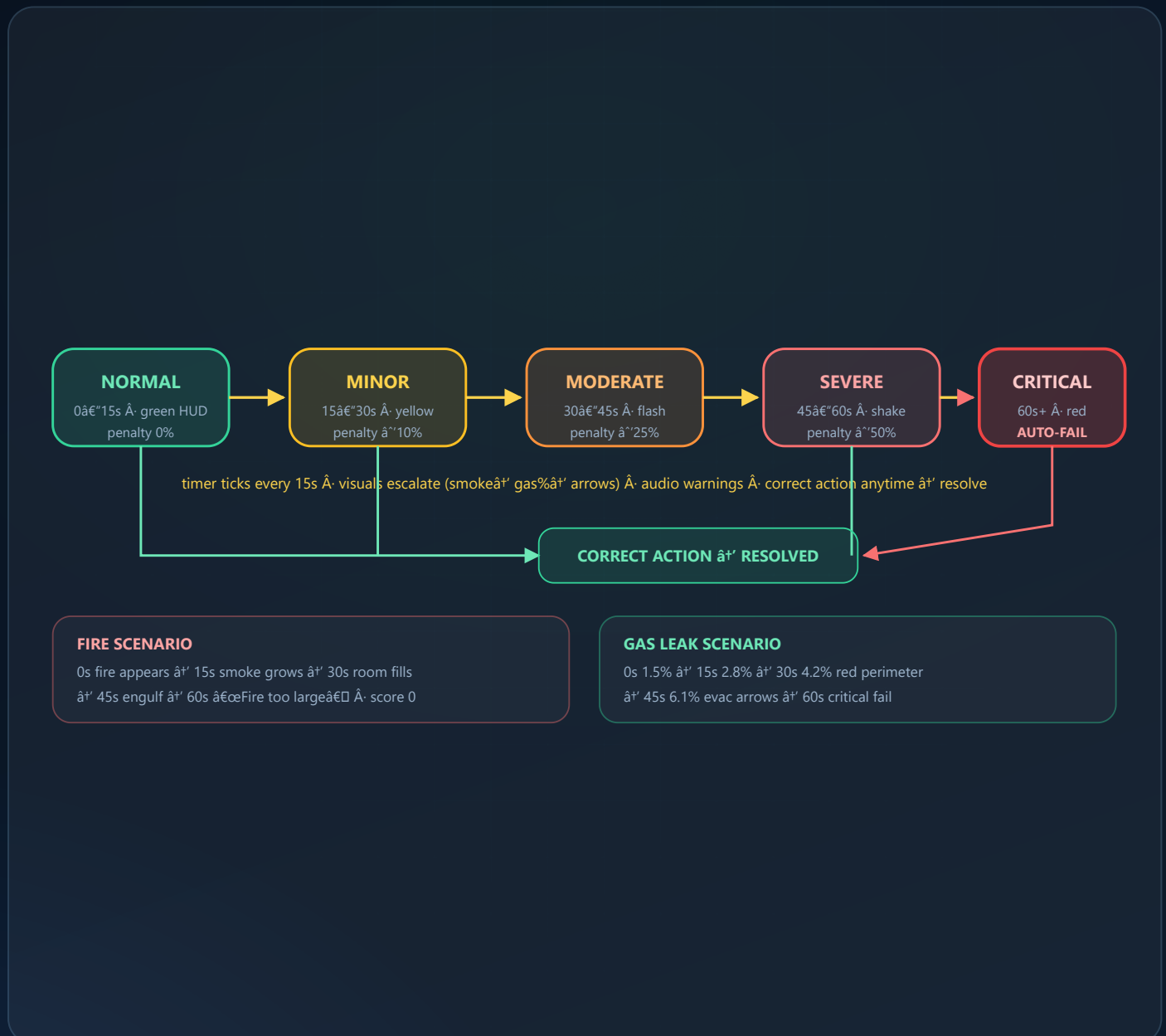
Training Session & Escalation Engine

THE HEART OF THE PRODUCT

A · AR TRAINING SESSION WORKFLOW

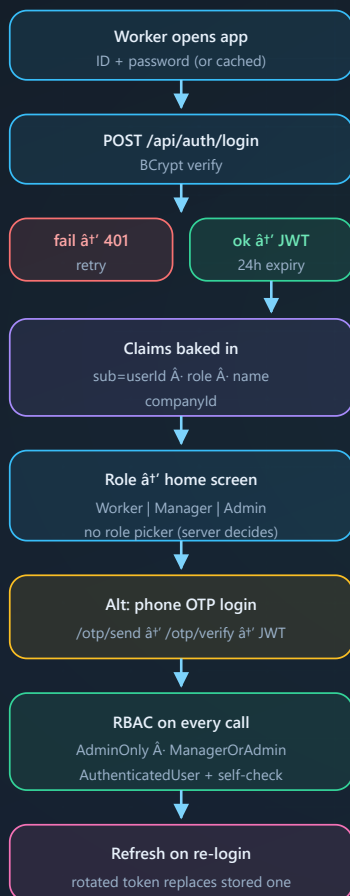


B · ESCALATION STATE MACHINE · INACTION WORSENS REALITY

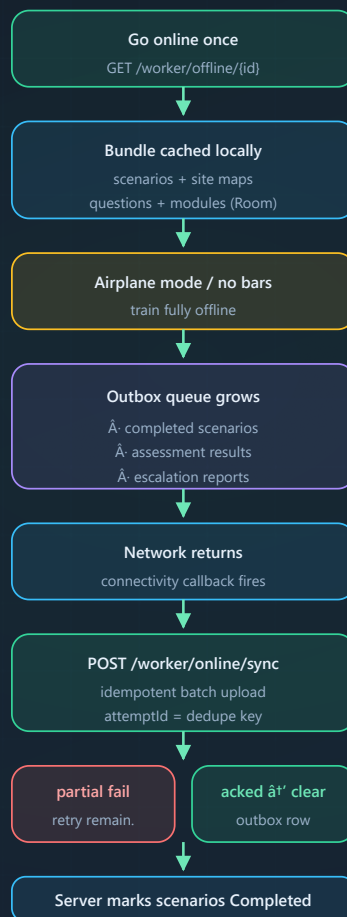


Auth & Offline Sync & Certificate

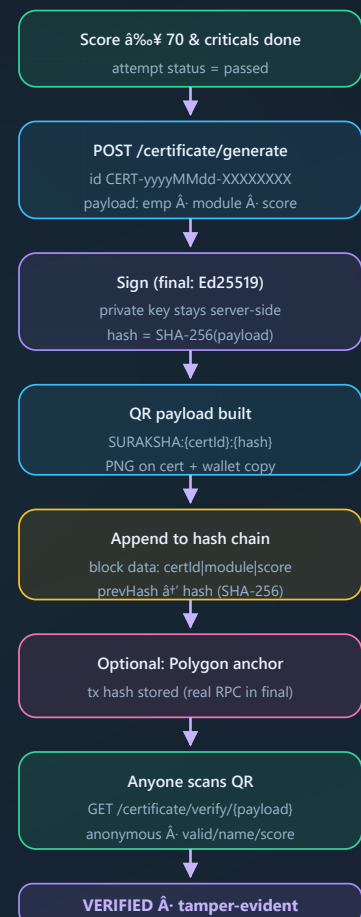
A AUTHENTICATION



B OFFLINE-FIRST SYNC

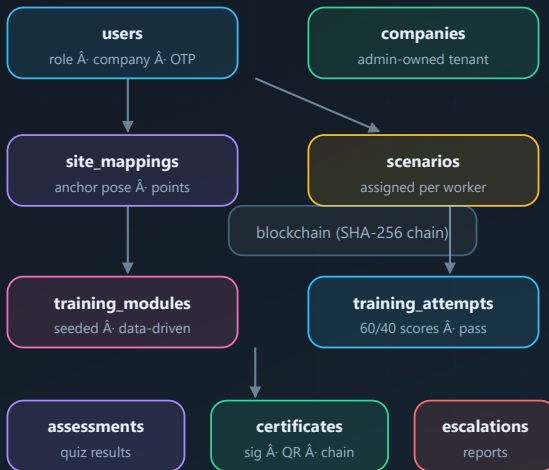


C CERTIFICATE LIFECYCLE



Collections, Contracts & Endpoint Map

MONGODB " 10 COLLECTIONS



Indexes on `UserId`, `CertificateId`, `AttemptId`, `AssessmentId`, `ModuleId`. camelCase convention. Raw BsonDocument at runtime for module/assessment flows.

REST API SURFACE (BY ROLE)

GROUP	ROLE	KEY ROUTES
Auth	public/self	login · otp/send · otp/verify · admin/seed · user/{id} · theme · language
Admin	AdminOnly	manager/add · worker/confirm · workers · sites · escalations · admins
Manager	Mgr Admin	worker/add · my workers · scenario/assign · stats · citification
Worker	auth	scenarios · escalation/report · site/record
Training	auth	modules · start · complete · history
Assessment	auth	questions · submit (final: answers stay server-side)
Certificate	auth/anon	generate · {id} · verify/{qr} · blockchain-anchor
Offline	auth	worker/offline/{id} · worker/online/sync
Ops	public	GET /health · Swagger (dev) · static /admin SPA

CROSS-CUTTING CONTRACTS

- Every authed call: **Authorization: Bearer <jwt>**
- Errors: 400 message · 401 unauth · 403 policy · 404 missing
- IDs are strings (GUIDs); dates ISO-8601 UTC
- Company scoping via **companyId** claim on list routes
- Workers can only read self (object-level checks)

SCORING CONTRACT

Total = **0.6** — action + **0.4** — quiz. Pass ≥ 70 (gas module 75). Escalation penalties applied to action score. Clamp 0–100, escalation level 0–4.

OFFLINE BUNDLE SHAPE

OfflineData = scenarios[] + siteMappings[] + lastSyncTime. Sync request = completedScenarios[] + escalations[], keyed by attemptId for idempotency.

CERTIFICATE PAYLOAD

certId · employeeId/name · moduleId/name · score · passed · issuedAt · signature · blockchainTx · qrPayload = **SURAKSHA:{id}:{hash}**

Roles, Screens & Feature Scope

WHO SEES WHAT

WORKER

Train · Prove · Repeat

- Home: today's training, X/5 progress
- QR scan → AR session
- HUD, timer, evacuation arrows
- Assessment + score breakdown
- Certificate wallet + QR
- Offline badge & sync status
- Language + voice settings

MANAGER

Roster · Assign · Verify

- Worker list + compliance %
- Add/remove own workers
- Assign scenarios to workers
- Certificate QR verification
- Escalation inbox (scoped)
- Site certification view
- Stats: total/active/pending

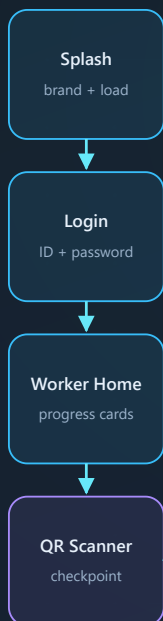
ADMIN (WEB)

Control · Audit · Scale

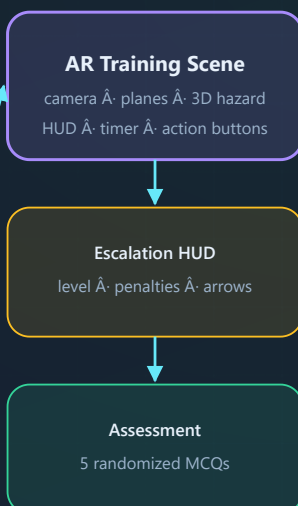
- System overview + stat cards
- Create managers & admins
- Confirm pending workers
- Site mappings management
- Escalation assign/resolve
- Module config (final scope)
- Company tenant isolation

SCREEN MAP · FINAL APP

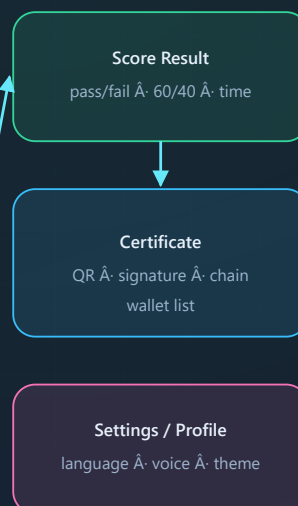
WORKER FLOW



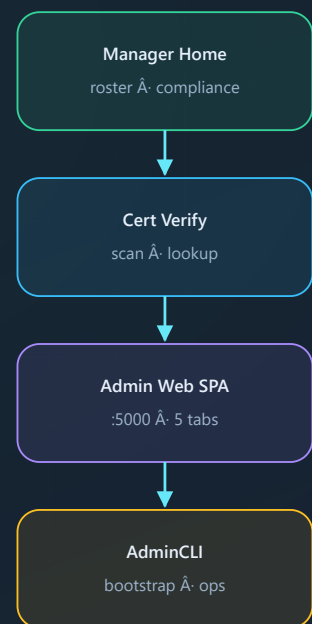
AR CORE



RESULT & PROOF



MANAGER / ADMIN



MUST (DEMO)

2 AR modules · assessment
· QR cert · Hindi+Santali
· offline train · admin web

SHOULD

Real outbox sync · Ed25519
· 3D assets · particles ·
voice 7 langs · TalkBack

COULD

Polygon anchor live ·
module config UI ·
manager analytics charts ·
Site view

WON'T (V1)

iOS · VR headset · live
multi-user drills · payments
· marketplace

Current State → Final State

HONEST BASELINE

WHAT EXISTS & WORKS TODAY

Backend API	works
JWT + RBAC	works
Mongo persistence	works
Admin web SPA	works
Cert QR + verify	works
Training modules seed	2 modules
Android screens (10)	works
Site-map AR (anchors)	works
Escalation reporting	works
EN/HI/SAT strings	48 keys
Local hash chain	works
Unity training scripts	reference

WHAT THE FINAL BUILD MUST ADD / FIX

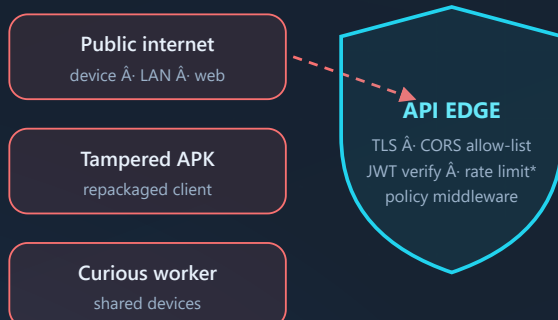
Real AR training scene	critical
Escalation engine live	critical
Offline outbox sync	critical
Single client decision	critical
Ed25519 real signing	security
Secrets out of repo	security
Quiz answers server-side	cheat-proof
CORS lock-down	security
Manager escalations bug	fix
Compile break (citification)	fix
Room DB + outbox	new
3D hazard assets	content

KNOWN ISSUES CARRIED INTO PLAN

- JWT secret hardcoded in 3 files + appsettings
- Assessment questions include correctAnswerIndex
- OTP never sent (Twilio unconfigured) → in-memory store
- POST /auth/admin/reset wipes entire users collection
- Dual frontend (Kotlin + Unity) model drift risk
- AdminCLI seed missing companyName → 400

SECURITY ARCHITECTURE → FINAL HARDENING

UNTRUSTED



Never trust the client: all scoring, issuance, and RBAC enforced server-side.

Identity

BCrypt hashes → short-lived JWT → OTP second factor (Twilio)

Authorization

Role policies → company claim scoping → self-or-admin object checks

Data integrity

Ed25519 cert sigs → hash chain → server-side scoring → audit log*

* env-var secrets → CORS origins → request rate limiting added in final

Implementation Roadmap “Final Build”

5 PHASES · ~6 WEEKS

PHASE TIMELINE

P1 · Stabilize



Wk 1

fix breaks · security
secrets · CORS · bugs

P2 · AR Core



Wk 2&3

fire + gas scenes
HUD · escalation live

P3 · Offline Sync



Wk 3&4

Room + outbox
idempotent upload

P4 · Trust Layer



Wk 4&5

Ed25519 · server quiz
Polygon · verify UX

P5 · Polish



Wk 5&6

3D art · i18n 7
APK · video · demo

Each phase ends with a runnable demo build “no big-bang integration.”

PHASE DELIVERABLES

PHASE	OUTCOME
P1	Green build · citification fix · env secrets · CORS list · manager escalation scope · AdminCLI seed fix
P2	Native AR fire + gas sessions with HUD, timer, escalation states, action taps, evacuation arrows
P3	Room cache of modules/sites · outbox queue · reconnect flush · airplane-mode demo path
P4	Ed25519 keypair · sign on issue · verify offline + online · quiz scoring only on server
P5	3D props + particles · 7-language pass · accessibility · signed APK · demo video · GitHub README

DEFINITION OF DONE (PRODUCT)

- Worker completes fire module fully offline, syncs later
- Escalation visibly worsens scene within 15s ticks
- Certificate QR verifies on a second phone, no login
- Manager sees only own company workers & escalations
- Admin dashboard confirms worker + assigns escalation
- Hindi & Santali UI + voice on critical prompts

SUCCESS METRICS

<60s

MEDIAN ACTION TIME

â‰¥70%

FIRST-PASS RATE

0

DATA LOSS OFFLINE

2+

LIVE AR MODULES

TOP RISKS & MITIGATIONS

Low-end AR perf â†’ budget mode, primitive+texture fallback, SceneView LOD.**Dual-client confusion â†’** freeze Unity as design reference; native is ship path.**Offline conflicts â†’** attemptId idempotency; last-writer only on non-critical fields.

— CLOSING — THE ASK

Ship one honest, offline-first AR trainer that mines will actually use.

BUILD

One Kotlin client with a real AR training scene, a hardened .NET API, and a Room-backed sync outbox — the architecture already sketched in this document.

PROVE

Scored sessions, escalation pressure, and certificates anyone can verify by QR — with the trust story (Ed25519 + chain) made real, not simulated.

DELIVER

SIH package: working APK, demo video, GitHub repo, admin dashboard — meeting every minimum requirement of problem statement 26041.

Real Hazards

Real Decisions

Safer Tomorrow

DOCUMENT CONTROL

SurakshaAR — Final Version Proposal — v1.0 — September 2026

Prepared from full review of E:\SAFE (Backend, Shared, SurakshaAR-Android, UnityAR, AdminCLI, REF, PROJECT_VISION, WORKFLOW)

Problem Statement SIH 26041 — Government of Jharkhand — Smart Education

